

18. (b) The first-generation small modular reactors are expected to use lowenriched uranium in facilities assembled on-site with factory-made parts, to produce waste that can be handled using existing technologies and power that can be sold at economical rates.

19. (a) Between the eighth and eleventh centuries A. D., the Byzantine Empire staged an almost unparalleled economic and cultural revival, a recovery that is all the more striking because it followed a long period of severe internal decline.

20. (b) In the two decades between 1910 and 1930, over ten percent of the Black population of the United States left the South, where the preponderance of the Black population had been located, and migrated to northern states, with the largest number moving, it is claimed, between 1916 and 1918.

21. (b) Hollow : without real value/Without significance. So (b) is the right option.

Momentary victory is worthless/meaningless.
consequential : important; that will have important results.
void - completely without something.
spatial - relating to space and the position, size, shape, etc. of things in it.

22. (b) Without - not having.

'Without' is itself a negative preposition so it has got a negative meaning. In negative sentences, non assertive words are used.

So 'anything' is the right option.

23. (b) 'It' is a singular subject so it will take a singular verb. On this basis option A got eliminated.

Main clause is in the present tense so it will use the present tense.

Has had - present perfect.

Had had - past perfect.

Had - past simple

24. (b) "Paint the town red" means to go out and enjoy yourself by drinking alcohol, dancing, laughing with friends, etc.

25. (c) Cool one's heels – **to be kept waiting**

26. (b) **cut someone some slack-** to give someone additional freedom

The idiom "cut us some slack" means **to be lenient or give someone a break**. Given that it was the last day of class, it makes sense that the teachers might have been more lenient with the students.

The other options do not fit the context or convey the correct meaning:

"Cut us down to size" means to humble someone.

"Cut us out of the class" suggests exclusion.

"Cut us short" implies to stop someone from talking before they have finished what they were saying

27. (d) The relationship between "Window" and "Pane" is that a window typically has panes (a framed sheet of glass in a window or door). Similarly, a book consists of "pages," which is the primary component of a book.

28. (b) Both Pernicious and Malignant carry a sense of extreme harm or evil; they describe something extremely detrimental or dangerous.

Likewise, Noxious and Nasty are words that denote extreme unpleasantness or harmfulness.

Noxious signifies something physically harmful or poisonous, while Poisonous implies causing or capable of causing death.

Let's see the other given options:

Option (a) (Harmful : Helpful): These words represent direct opposites, but they don't reflect the extreme, malevolent nature that Pernicious and Malignant carry.

Option (c) (Destructive : Constructive): These words represent opposite actions or qualities, but they don't capture the malevolent and extremely harmful sense of Pernicious and Malignant.

Option (d) (Detrimental : Advantageous): These words represent opposing effects, but they don't mirror the strong negative connotations and extreme harm indicated by Pernicious and Malignant.

29. (b) **Gerascophobia is the fear of aging.**

Claustrophobia is the fear of confined spaces.

Thanatophobia is the fear of death.

Acrophobia is the fear of heights.

Hence, the correct term for the fear of aging is "Gerascophobia."

30. (b) **Altruist** means A lover of mankind.

Synonym: Philanthropist

Antonym: Misanthrope

31. (a) Let the two numbers be x and y .

Here

$$(x + y)^2 = 90 \quad \dots (i)$$

$$(x - y)^2 = 46 \quad \dots (ii)$$

Subtracting eq. (ii) from eq. (i)

$$(x + y)^2 - (x - y)^2 = 90 - 46$$

$$\Rightarrow 4xy = 44$$

$$\Rightarrow xy = 11$$

32. (b) Let the rate of interest be R .

$$\Rightarrow 1348.32 = 1200 \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \left(1 + \frac{R}{100} \right)^2 = \frac{1348.32}{1200}$$

$$\Rightarrow \left(1 + \frac{R}{100} \right)^2 = \frac{11236}{10000}$$

$$\Rightarrow 1 + \frac{R}{100} = \frac{106}{100}$$

$$\Rightarrow R = 6\%$$

33. (d) Let CP be 100.

Profit = 30%

$$\therefore SP = 130$$

$$\text{Now, profit on selling price} = \left(\frac{30}{130} \right) \times 100$$

$$= 23.07\%$$

34. (c) Let the distance be D km.

$$\text{Speed} = \frac{\text{distance}}{\text{time}} = \frac{D}{\left(\frac{45}{60} \right)} = \frac{60D}{45}$$

$$= \frac{4D}{3} \text{ km/h}$$

$$\text{New speed} = \left(\frac{4D}{3} \right) - 5 = \frac{(4D - 15)}{3} \text{ km/h}$$

$$\text{New time} = 45 \text{ min} + 3 \text{ min} = 48 \text{ min}$$

$$\frac{48}{60} \times \frac{(4D - 15)}{3} = D$$

$$\Rightarrow 16(4D - 15) = 60D$$

$$\Rightarrow 4(4D - 15) = 15D$$

$$\Rightarrow 16D - 60 = 15D$$

$$\Rightarrow D = 60$$

35. (c) $\frac{a}{b} = \frac{3}{5}$

$$\Rightarrow a = \frac{3b}{5}$$

$$\Rightarrow 2a = \frac{6b}{5} \quad \dots (i)$$

$$\frac{b}{c} = \frac{7}{8}$$

$$\Rightarrow 7c = 8b \quad \dots (ii)$$

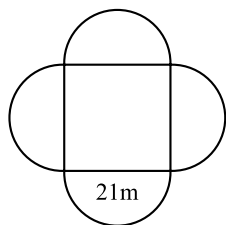
Using eqn. (i) and (ii) in $(2a : 3b : 7c)$

$$\Rightarrow \frac{6b}{5} : 3b : 8b$$

$$\Rightarrow 6 : 15 : 40$$

36. (a) Length of the fence

$$= 4\pi R \text{ where, } R = \frac{21}{2} m$$



$$\Rightarrow 4\pi R \Rightarrow 4 \times \frac{22}{7} \times \frac{21}{2} m$$

$$= 132 m$$

$$\text{Cost of fencing} = ₹ \left(132 \times \frac{25}{2} \right) = ₹ 1650$$

37. (d) Sum of ages of five persons = $28 \times 5 = 140$

$$\text{Sum of ages of four persons} = 26 \times 4 = 104$$

$$\text{The age of the excluded person}$$

$$= 140 - 104 = 36 \text{ years.}$$

38. (a) $CP = 30 \times \frac{100}{150} = ₹ 20$ per kg

$$\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}}$$

$$= \frac{\text{Dearer price} - \text{Mean price}}{\text{Mean price} - \text{Cheaper price}}$$

$$\text{Required ratio} = \frac{24.8 - 20}{20 - 18} = \frac{12}{5} = 12 : 5$$

39. (a) Let SP of 1 kg rice = ₹ 1

$$\text{CP of 80 kg rice} = 80 \times \frac{90}{100} = ₹ 72$$

$$\text{Rice got free on 80 kg} = \frac{80}{20} = 4 \text{ kg}$$

$$\text{Total rice} = 80 + 4 = 84 \text{ kg}$$

$$\text{Total SP of total rice} = ₹ 84$$

$$\text{Profit} = 84 - 72 = ₹ 12$$

$$\text{Profit percentage} = \left[\frac{12}{72} \right] \times 100 = 16 \frac{2}{3} \%$$

40. (c) Rate = 4% per annum which is 2% per half-yearly

$$\text{Time} = 1 \text{ year} = 2 \text{ half-years}$$

$$A = P \left(1 + \frac{R}{100} \right)^t$$

$$A = 8000 \left(1 + \frac{2}{100} \right)^2$$

$$= 8000 \left(\frac{51}{50} \right)^2$$

$$= ₹ 8323.20$$

$$\text{Interest} = A - P = 8323.20 - 8000$$

$$= ₹ 323.20$$

41. (c) Let the amount borrowed = p

$$\text{Then, } 405 = \frac{p \times 5 \times 4}{100} + \frac{p \times 4 \times 4}{100}$$

$$\Rightarrow 40500 = 36p$$

$$\Rightarrow p = ₹ 1125$$

42. (d) Smallest 5-digit number = 10000

On dividing 10000 by 243, we get

$$\text{Remainder} = 37$$

$$\text{So, } 243 - 37 = 206$$

206 should be added to 10000 to get the smallest 5-digit number divisible by 243.

$$\text{Required number} = 10000 + 206 = 10206$$

43. (b) Let the initial price of cooking gas be ₹ 100

$$\text{Price after increase} = ₹ 115$$

On ₹ 115 he should reduce ₹ 15.

Percentage decrease in consumption

$$= \frac{15}{115} \times 100 = 13 \frac{1}{23} \%$$

Short method:

If price of a commodity is increased (or decreased) by $x\%$, then the decrease (or increase) in consumption, so as not to increase (or decrease) the expenditure is

$$\left(\frac{x}{100 \pm x} \times 100 \right) \%$$

Here $x = 15\%$

Reduction in consumption

$$= \left(\frac{x}{100 + x} \times 100 \right) \%$$

$$= \frac{15}{115} \times 100\% = 13 \frac{1}{23} \%$$

44. (c) (A's one day's work) : (B's one day's work)

$$= 150 : 100 = 3 : 2$$

Let A's one day's work be $3x$ and B's one day's work be $2x$.

$$\therefore \text{C's one day's work} = \frac{(3x + 2x)}{2} = \frac{5x}{2}$$

$$\text{Now, } \frac{5x}{2} = \frac{1}{40}$$

$$\Rightarrow x = \left(\frac{1}{40} \right) \times \frac{2}{5}$$

$$\Rightarrow x = \frac{1}{100}$$

$\therefore (A + B + C)$'s one day's work:

$$\Rightarrow \frac{3}{100} + \frac{1}{50} + \frac{1}{40}$$

$$\Rightarrow \frac{15}{200}$$

$$\Rightarrow \frac{3}{40} = (A + B + C)\text{'s one day work}$$

A, B, C can complete the whole work in $\frac{40}{3}$ days.

45. (a) Interest rate for 8 monthly = $15 \times \frac{8}{12}$

$$= 10\%$$

Let the share of Anuj be x .

As per the question,

$$x \left(1 + \frac{10}{100} \right)^3 = (8400 - x) \left(1 + \frac{10}{100} \right)^4$$

$$\Rightarrow x = (8400 - x) \times \frac{11}{10}$$

$$10x = 11 \times 8400 - 11x$$

$$\Rightarrow 21x = 11 \times 8400$$

$$\Rightarrow ₹ 4400$$

Anuj's share in the given sum is ₹ 4400

46. (b) Total number of events = ${}^{12}C_2$

$$\frac{(12!)}{(2! \times 10!)}$$

$$= 6 \times 11 = 66$$

Number of favourable events

$$= {}^4C_2 = \frac{(4!)}{(2! \times 2!)}$$

$$= \frac{(4 \times 3)}{2} = 6$$

Probability of getting exactly two rotten apples

$$= \frac{6}{66} = \frac{1}{11}$$

47. (a) Revenue = price $\times$ sales

Let the original price per item be ₹ 100.

and the number of items sold be 100.

Then, original revenue = 100×100

$$= ₹ 10000.$$

New price = $(100 - 20) \%$ of 100 = ₹ 80

New revenue = $(100 + 25) \%$ of 10000

$$= ₹ 12500$$

New number of items sold

$$= \frac{12500}{80} = 156.25 = 156.25$$

Percentage increase in the number of items sold

$$= \frac{156.25 - 100}{100} \times 100 = 56.25\%$$

Hence, the value of $x = 56.25$

48. (b) A can finish the work in 25 days.

So, in one day A can do $\left(\frac{100}{25} \right) \% = 4\%$ of the work.